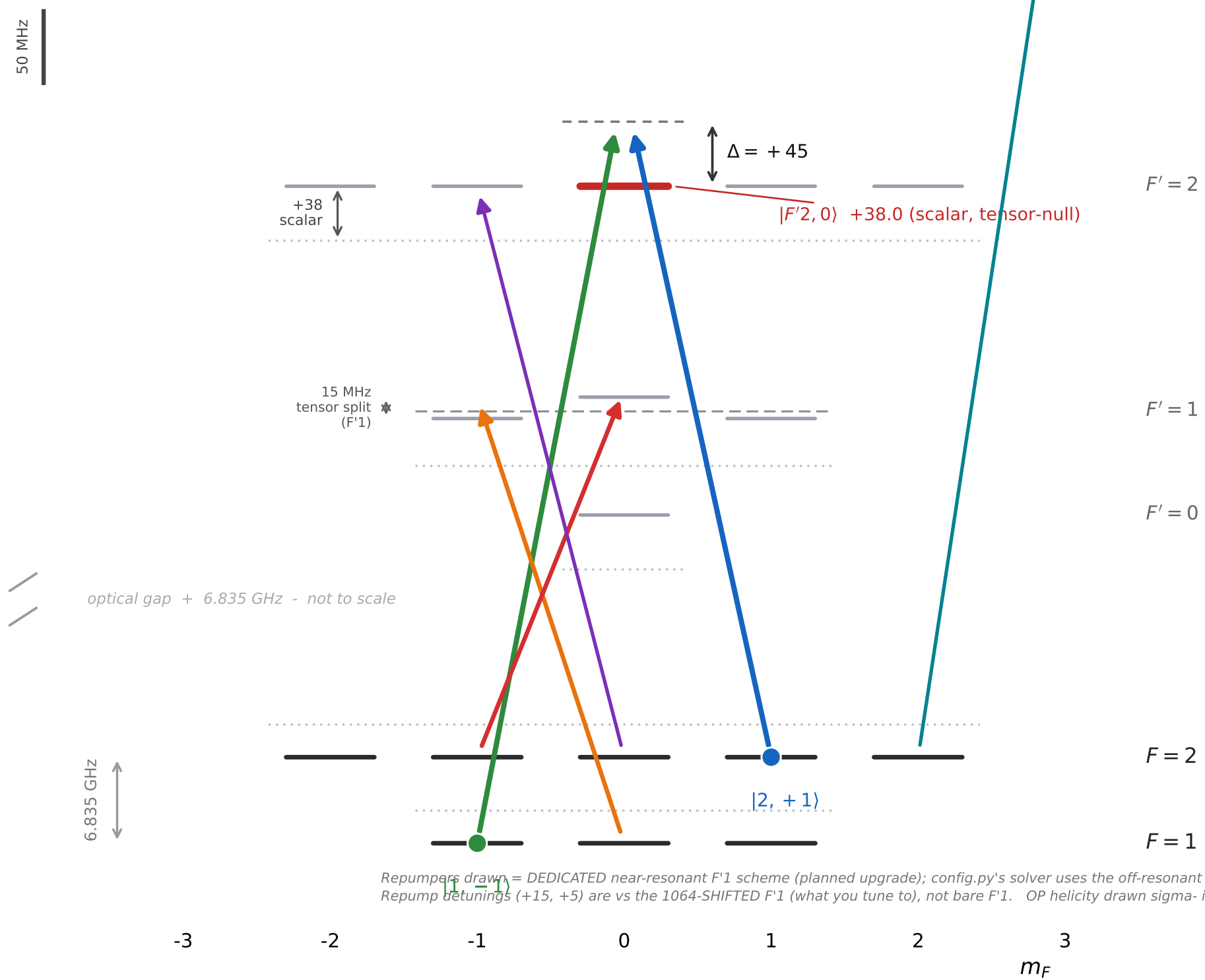


⁸⁷Rb D2 clock-EIT cooling - the six beams on the 1064-shifted 5P_{3/2} manifold (grey = un-shifted reference)

- grey reference lines
- bare - no 1064 shift
- - - scalar only (+38, no tensor)
- full = scalar + tensor (solid)

- beams (EIT Λ + two σ repumpers + master prep/readout)
- control σ^- . F2 $\rightarrow$ F'2 . $\Delta = +45$ above $|F'2, 0\rangle$. $\Omega/2\pi = 8.8$ MHz
 - probe σ^+ . F1 $\rightarrow$ F'2 . $\Delta = +45$. $\Omega/2\pi = 1.1$ MHz ($\Omega_p/\Omega_c = 0.12$)
 - repump1 σ^- . F1 $\rightarrow$ F'1 . +15 above F'1 . $\Omega/2\pi \approx 3$ MHz
 - repump2 σ^+ . F2 $\rightarrow$ F'1 . +5 above F'1 . $\Omega/2\pi \approx 3$ MHz
 - OP σ . F2 $\rightarrow$ F'2 . on-resonance . saturating (state prep)
 - detection σ^+ . F2 $\rightarrow$ F'3 . on-resonance . cycling (readout)



1064 nm light shifts (atom at trap centre, theta=90 deg)
ground 5S1/2: -22.7 scalar - F=1 & F=2 together (6.835 GHz kept)
excited 5P3/2: +38.0 scalar (common) + tensor:
|F'2,m'> = +38.0 (tensor null - clean target, any geometry)
|F'1,+/-1> = +33.0 , |F'1,0> = +47.9 (15 MHz split)
|F'3,+/-3> = +47.3 -> |F'3,0> = +30.5 (17 MHz spread)
|F'0,0> = +38.0 (tensor null)
=> F2->|F'2,0> sits +60.7 vs bare (+38 up, ground -22.7 down);
probe & control are +45 above that.
=> split F'1 => repump sigma+/- legs see unequal detunings
(the F'=1 inhomogeneity). [all values 2*pi*MHz]

Repumpers drawn = DEDICATED near-resonant F'1 scheme (planned upgrade); config.py's solver uses the off-resonant single-EOM comb (rep1 F1->F'2 445-off, rep2 F2->F'1 198-off).
Repump detunings (+15, +5) are vs the 1064-SHIFTED F'1 (what you tune to), not bare F'1. OP helicity drawn sigma- is a placeholder - confirm against the real state-prep sequence.